

Student Registration System

DATA MODEL & DATA DICTIONARY

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1. Purpose

This document defines the core business data entities, relationships, logical attributes, keys, status values, and data rules required to support the Student Registration System. It is intended as a Business Analysis logical data artifact rather than a physical database design.

2. Scope

The model covers student administration, academic terms and offerings, registration and enrollment, waitlists, billing, payments, financial aid, identity/authorization references, notifications, and audit information.

3. Conceptual Data Model

Core conceptual relationships:

Student 1 — M Enrollment M — 1 Course Section

Course 1 — M Course Section

Academic Term 1 — M Course Section
 Student 1 — M Waitlist Entry M — 1 Course Section
 Student 1 — M Invoice 1 — M Invoice Line
 Invoice 1 — M Payment Allocation M — 1 Payment
 Student 1 — M Financial Aid Award
 User M — M Role (resolved through User Role)
 Student 1 — M Notification
 User / System Event 1 — M Audit Event

4. Entity Relationship Overview

Parent Entity	Cardinality	Child Entity	Business Meaning
Student	1:M	Enrollment	A student may have many enrollment records.
Course	1:M	Course Section	A course may be offered through many sections.
Academic Term	1:M	Course Section	A term contains many course-section offerings.
Course Section	1:M	Enrollment	A section may contain many enrolled students.
Student	1:M	Waitlist Entry	A student may join multiple section waitlists.
Course Section	1:M	Waitlist Entry	A section may have multiple waitlisted students.
Student	1:M	Invoice	A student may receive multiple invoices over time.
Invoice	1:M	Invoice Line	An invoice contains one or more charge lines.
Student	1:M	Payment	A student may make multiple payments.
Invoice	M:N	Payment	Payments may be allocated to invoices through Payment Allocation.
Student	1:M	Financial Aid Award	A student may receive multiple aid awards.
User	M:N	Role	Users may have roles; roles may be assigned to multiple users.
Student	1:M	Notification	A student may receive many notifications.

5. Logical Entity Catalog

Entity	Business Definition	Primary Key
Student	Represents the university student and registration-relevant academic status.	Student_ID
User Account	Represents an authenticated system identity.	User_ID
Role	Represents an authorization role.	Role_ID
User Role	Associates users with assigned roles.	User_Role_ID

Academic Term	Represents an academic semester/term and registration dates.	Term_ID
Course	Represents a catalog-level academic course.	Course_ID
Course Section	Represents a scheduled offering of a course in a term.	Section_ID
Enrollment	Represents a student's registration relationship with a course section.	Enrollment_ID
Waitlist Entry	Represents a student's position/status on a section waitlist.	Waitlist_ID
Registration Hold	Represents a restriction that may block or condition registration.	Hold_ID
Invoice	Represents a billed financial obligation for a student.	Invoice_ID
Invoice Line	Represents an individual tuition/fee charge on an invoice.	Invoice_Line_ID
Payment	Represents a student payment transaction.	Payment_ID
Payment Allocation	Associates a payment amount with an invoice.	Allocation_ID
Financial Aid Award	Represents a scholarship, grant, waiver, or aid award.	Aid_Award_ID
Notification	Represents a system communication to a user/student.	Notification_ID
Audit Event	Represents an auditable security/business event.	Audit_Event_ID

6. Data Dictionary

Student

Attribute	Data Type	Key / Constraint	Nullable?	Business Definition
Student_ID	UUID / Integer	PK	No	Unique student identifier
University_Email	String	Unique	No	Approved university email
First_Name	String		No	Student first name
Last_Name	String		No	Student last name
Academic_Status	Enum		No	Current academic standing/status
Program_Code	String	FK/Reference	Yes	Academic program reference

Academic Term

Attribute	Data Type	Key / Constraint	Nullable?	Business Definition
Term_ID	UUID / Integer	PK	No	Unique term identifier
Term_Code	String	Unique	No	Business term code
Term_Name	String		No	Display name
Start_Date	Date		No	Term start date
End_Date	Date		No	Term end date
Registration_Open	DateTime		No	Registration opening

Registration_Close	DateTime		No	Registration closing
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Course

Attribute	Data Type	Key / Constraint	Nullable?	Business Definition
Course_ID	UUID / Integer	PK	No	Unique course identifier
Course_Code	String	Unique	No	Catalog course code
Course_Title	String		No	Course title
Credit_Hours	Decimal		No	Default credit hours
Department_Code	String	Reference	No	Owning academic department

Course Section

Attribute	Data Type	Key / Constraint	Nullable?	Business Definition
Section_ID	UUID / Integer	PK	No	Unique section identifier
Course_ID	UUID / Integer	FK	No	Related course
Term_ID	UUID / Integer	FK	No	Related academic term
Section_Number	String		No	Section number
Capacity	Integer		No	Maximum enrollment capacity
Enrolled_Count	Integer		No	Current confirmed enrollment count
Section_Status	Enum		No	Open, Closed, Cancelled, etc.
Meeting_Pattern	String		Yes	Scheduled meeting pattern

Enrollment

Attribute	Data Type	Key / Constraint	Nullable?	Business Definition
Enrollment_ID	UUID / Integer	PK	No	Unique enrollment identifier
Student_ID	UUID / Integer	FK	No	Student
Section_ID	UUID / Integer	FK	No	Course section
Enrollment_Status	Enum		No	Pending, Enrolled, Dropped, Withdrawn, Rejected
Enrollment_Date	DateTime		No	Registration transaction timestamp
Credit_Hours	Decimal		No	Credits applied to this enrollment

Waitlist Entry

Attribute	Data Type	Key / Constraint	Nullable?	Business Definition
Waitlist_ID	UUID / Integer	PK	No	Unique waitlist entry
Student_ID	UUID / Integer	FK	No	Student
Section_ID	UUID / Integer	FK	No	Course section
Position	Integer		No	Current waitlist priority/position

Waitlist_Status	Enum		No	Active, Offered, Promoted, Expired, Removed
Joined_At	DateTime		No	Waitlist entry timestamp

Invoice

Attribute	Data Type	Key / Constraint	Nullable?	Business Definition
Invoice_ID	UUID / Integer	PK	No	Unique invoice identifier
Student_ID	UUID / Integer	FK	No	Student
Term_ID	UUID / Integer	FK	No	Related term
Invoice_Status	Enum		No	Draft, Issued, Partially Paid, Paid, Cancelled
Total_Amount	Decimal		No	Invoice total
Amount_Due	Decimal		No	Outstanding amount

Payment

Attribute	Data Type	Key / Constraint	Nullable?	Business Definition
Payment_ID	UUID / Integer	PK	No	Unique payment identifier
Student_ID	UUID / Integer	FK	No	Payer/student
Amount	Decimal		No	Payment amount
Payment_Status	Enum		No	Pending, Authorized, Successful, Failed, Refunded, Reconciled
Gateway_Reference	String	External Ref	Yes	Payment-provider reference
Paid_At	DateTime		Yes	Successful payment timestamp

Financial Aid Award

Attribute	Data Type	Key / Constraint	Nullable?	Business Definition
Aid_Award_ID	UUID / Integer	PK	No	Unique aid award
Student_ID	UUID / Integer	FK	No	Recipient student
Aid_Type	Enum		No	Grant, Scholarship, Waiver, Discount
Award_Amount	Decimal		No	Approved amount
Award_Status	Enum		No	Pending, Approved, Applied, Cancelled
Term_ID	UUID / Integer	FK	No	Applicable academic term

7. Key Status / Reference Values

Attribute	Allowed / Illustrative Values
Enrollment_Status	Pending, Enrolled, Waitlisted, Dropped, Withdrawn, Rejected
Waitlist_Status	Active, Offered, Promoted, Expired, Removed
Section_Status	Draft, Open, Closed, Full, Cancelled
Invoice_Status	Draft, Issued, Partially Paid, Paid, Cancelled

Payment_Status	Pending, Authorized, Successful, Failed, Refunded, Reconciled
Aid_Award_Status	Pending, Approved, Applied, Cancelled

8. Core Data Business Rules

Each active Enrollment must reference one valid Student and one valid Course Section.

A student must not have more than one active enrollment for the same Course Section.

Confirmed enrollment must not cause Enrolled_Count to exceed Section Capacity.

Seat allocation and enrollment confirmation must be transactionally consistent under concurrent requests.

An active Waitlist Entry must reference a section that permits waitlisting.

A Payment must not be treated as successful until the authoritative payment result confirms success.

Financial aid must be applied only when the award is approved and applicable to the relevant term/charges.

Registration status and student schedule data must remain synchronized after confirmed add, drop, or withdrawal actions.

Security-sensitive and financially significant changes must produce an Audit Event.

9. Data Ownership & Source Systems

Data Domain	Authoritative Source	Notes
Student profile / academic status	Student Information System (SIS)	Student Registration System consumes authoritative student data.
Course / section / term	Academic Operations / SIS	Published offerings are authoritative for registration.
Enrollment / waitlist	Student Registration System	SRS owns registration transaction state.
Invoice / payment status	Bursar / Financial System	Financial system is authoritative for account balances and reconciliation.
Financial aid	Financial Aid System / Office	Approved award source.
Identity / role	Identity & Access Management	Authoritative authentication/authorization source.
Audit events	Audit / Monitoring Service	Central audit record for in-scope events.